DCOR: Dynamic Channel-Wise Outlier Removal to De-Noise LiDAR Data Corrupted by Snow

Shanglian Zhou, Hao Xu, Guohui Zhang, Tianwei Ma, and Yin Yang

***Abstract*— Since the past decade, Light Detection and Ranging (LiDAR) data have been extensively adopted for traffic object recognition tasks. Existing methodologies often assume LiDAR data are acquired under normal weather conditions. Nevertheless, many researchers have observed that the LiDAR data captured under inclement weather are often contaminated with noises such as fog and snow, which may deteriorate the data quality and lead to false detections in traffic object recognition. This paper proposes a neighborhood-based noise removal methodology to eliminate snow noises from LiDAR data. It identifies a point of interest from a specific laser channel as an outlier, if the number of neighboring points in the same channel within a dynamic search radius is fewer than a threshold. Unlike existing methods that filter the entire LiDAR point cloud, the proposed methodology processes LiDAR data channel-by-channel, which helps reduce the data dimensionality and decouple the snow effects along the vertical axis of the 3D point cloud, leading to more effective and efficient outlier detection. Furthermore, by dynamically changing the search radius based on the point-to-sensor distance rather than adopting a fixed search radius, the proposed methodology can account for the reduced point density at far distances caused by the non-uniformity of LiDAR data. In the experimental study, the proposed methodology is compared against some existing LiDAR de-noising approaches, including two state-of-the-art methods, and demonstrates superior performance in both accuracy (i.e., F1 score = 98.3%) and efficiency.**

***Index Terms*— LiDAR data, adverse weather, snow noise, channel-wise outlier removal, dynamic search radius.**

I. Introduction

RECENT years have witnessed groundbreaking developments in the acquisition and processing of light Detection and Ranging (LiDAR) data for traffic monitoring, remote sensing, and object recognition tasks [1], [2], [3], [4]. For example, Autonomous Vehicles (AVs) often rely on LiDAR data for environment perception and object recognition [3]. Despite the exuberant growth of LiDAR sensing technologies and data processing techniques, several challenges related to this type of data are yet to be fully addressed.

One of the pressing concerns is the noise contamination in LiDAR data under severe weather conditions. Many researchers have observed that the LiDAR data acquired in adverse weather, such as wind, rain, snow, and fog, often suffer from deteriorated quality [5], [6], [7], [8], [9], [10], [11]. In a LiDAR point cloud, snow particles are often shown as diffuse, solid objects dispersing around the LiDAR sensor. Depending on the rate of precipitation and density of particles, it can be particularly challenging for LiDAR devices to operate normally under such weather situations because laser beams can be easily backscattered from these tiny particles [7].

To address the issue of noise contamination in LiDAR data, a popular approach in AVs is to leverage multi-modal data fusion, by integrating information from other sources such as radar data and RGB images to reduce the data uncertainty in each individual data source through cross-domain feature correlation [12], [13]. Nevertheless, the development and maintenance of multi-sensor platforms are often time-consuming and expensive. Alternatively, many researchers have developed and applied data pre-processing approaches to de-noise the LiDAR data captured under adverse weather conditions prior to feature extraction. Although two-dimensional (2D) or three-dimensional (3D) general-purpose de-noising methods such as median filtering [14] have been adopted to eliminate snow noises from LiDAR data, they often fail to deliver satisfactory performance in snow removal because they do not account for the non-uniformity of LiDAR data. More recently, a few methodologies specifically designed to address the issue of snow noises in LiDAR data have been reported in the literature, which showed promising results on snow removal by taking into consideration the unique characteristics of LiDAR point clouds and snow noises. Nevertheless, in the existing literature, research findings on de-noising LiDAR data corrupted by snow are still insufficient, and thus further efforts need to be devoted to developing and improving LiDAR de-noising methodologies in terms of their accuracy and efficiency in processing real-world LiDAR data captured under severe snowy weather conditions.

This paper proposes Dynamic Channel-wise Outlier Removal (DCOR), a neighborhood-based noise removal approach, to eliminate snow noises from LiDAR data. The proposed methodology processes LiDAR data based on each laser channel and identifies a point of interest as an outlier if

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the number of neighboring points in the same laser channel within a search radius is fewer than a threshold. The search radius for each point is dynamically adjusted based on the point-to-sensor distance, such that it can account for the varying point density that decreases as the distance to sensor increases. The technical contributions can be summarized as follows:

- The Dynamic Channel-wise Outlier Removal (DCOR) filter is proposed to de-noise LiDAR data corrupted by snow. Different than the existing state-of-the-art approaches, the proposed method processes LiDAR data channel-by-channel rather than filtering the entire point cloud, which helps reduce the data dimensionality and decouple the snow effects along the vertical axis. In the experimental study, the cross comparisons between DCOR and two state-of-the-art methods have demonstrated that the proposed methodology yields superior performance in both accuracy and efficiency.
- Upon elaborating the technical details of the proposed methodology, the physical meanings and impacts of the two parameters, namely the distance factor f and the minimum number of neighboring points *MinPts*, are discussed through theoretical derivations, thus providing insights and prior knowledge for future similar studies and applications.

The rest of this paper is organized as follows: the "Related Work" section first provides a literature review on the impact of severe snowy weather conditions and then introduces the development of noise removal methods for LiDAR de-noising purposes, in which four existing approaches implemented for cross comparisons are described in detail. The "Methodology" section first introduces the LiDAR working mechanism and the characteristics of LiDAR data and snow noises, and then describes the technical details of the proposed DCOR filter; subsequently, the physical meanings of the parameters involved in the proposed methodology are elaborated through theoretical derivations, and their impacts on the de-noising performance are discussed and assessed qualitatively. The "Experimental Study" section presents two experimental cases. Case I investigates the optimal choice of f and *MinPts* through grid search, and Case II further compares the proposed methodology against four existing approaches on snow removal. The last section "Conclusion" summarizes the research findings and discussions to provide insights and prior knowledge for future studies and applications.

II. Related Work

It has been widely observed and reported by researchers that severe weather conditions such as snow and fog can often deteriorate the quality of LiDAR data captured under such scenarios [5], [6], [7], [8], [9], [10], [11]. Yamauchi [5] observed that LiDAR devices can provide good accuracy and resolution in normal weather but have difficulties functioning under precipitation and low visibility conditions. Michaud et al. [6] performed an experimental study to characterize the behavior of LiDAR devices in snowy conditions. They discovered that the distribution of snowflakes is very close to a log-normal distribution, and the noises caused by snowflakes backscattering laser beams are often detectable within the distance of 10 meters to the sensor location. Charron et al. [8] reported that the amount of snow decreases when the distance to the sensor increases, with a maximum detectable range between 10 meters to 20 meters. Kutila et al. [7] tested the performance of two LiDAR devices using different laser wavelengths under stabilized foggy and rainy conditions in a weather chamber; they reported the LiDAR sensors suffered deteriorated performance when the adverse weather condition became more severe and decreased the visibility range. They also recommended a higher (i.e., 1550 nm) operating wavelength for LiDAR sensors, leading to less scattering on fog particles. Bijelic et al. [15] tested the performance of four LiDAR systems in controlled conditions in a fog chamber. They investigated the optimal configurations of the internal parameters of the LiDAR devices to improve their performance under adverse weather. Jokela et al. [9] performed a benchmark testing on some popular LiDAR sensors to compare their performance in adverse weather. They observed that all the tested LiDAR sensors performed worse in fog and snow than in clear weather conditions. As revealed by the study in [16], rainy weather can severely deteriorate the quality of LiDAR point clouds and lead to drastic performance degradation in LiDAR-based object detection methods.

From the perspective of feature representation and extraction, existing methodologies to de-noise LiDAR point clouds can be mainly categorized as 2D or 3D-based methods. By transforming the information stored in 3D LiDAR data into a 2D representation such as a depth map or an intensity map, traditional image processing methods such as 2D median filtering [14] and Gaussian low-pass filtering [14] can be readily adapted. However, as pointed out by Charron et al. [8], these methods primarily focus on smoothing noisy surface points and are not specifically designed to remove data outliers in an open 3D space, as is the case with LiDAR data corrupted by snow. As a result, 2D-based filtering methods are often unable to remove snow noises effectively due to the sparsity of LiDAR point clouds, and they have the negative effect of smoothing edges in key point features [8].

Compared to 2D-based methods for snow removal, 3D-based methods can be directly applied to processing 3D data and thus are better suited to the nature of LiDAR point clouds. The majority of existing 3D-based methods for noise removal are neighborhood-based. This type of methodology usually determines whether a point is an outlier based on the geometric and statistical properties (e.g., distance and distribution) of all similar points in the vicinity. Popular neighborhood-based methods include the Statistical Outlier Removal (SOR) filter [17], Radius Outlier Removal (ROR) filter [17], Density-based Spatial Clustering of Applications with Noise (DBSCAN) [18], etc. Nevertheless, these general-purpose noise removal methods often fail to deliver satisfactory performance on snow removal because they do not account for the non-uniformity of LiDAR point clouds. More recently, a few 3D de-noising methodologies specifically designed to process LiDAR data corrupted by snow have been reported in the literature and demonstrated promising results.

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Charron et al. [8] developed the Dynamic Radius Outlier Removal (DROR) filter for snow removal by adjusting the search radius based on the point-to-sensor distance, which takes into consideration the non-uniform distribution of LiDAR point clouds. Wu et al. [19] proposed 3D-SDBSCAN by revising the conventional DBSCAN algorithm to divide a 3D LiDAR space into two subareas based on the distribution of snowflakes and employ different DBSCAN parameters in different subareas. Park et al. [20] proposed the Low-intensity Outlier Removal (LIOR) filter, an intensity-based approach, to eliminate snow noises by deleting points with intensity values below a specified intensity threshold rather than detecting outliers based on the distance between points. By using LiLaNet [21] as the backbone, Heinzler et al. [22] proposed WeatherNet, a deep learning-based LiDAR de-noising method to reduce noises caused by severe weather conditions, such as rain, snow, and fog. Kurup and Bos [11] extended the SOR filter to account for the non-uniformity in LiDAR data by introducing a distance multiplication factor. Roriz et al. [23] developed a LiDAR de-noising method called Dynamic light-Intensity Outlier Removal (DIOR), which combines DROR and LIOR for snow removal and leverages the Field-Programmable Gate Array (FPGA) technology for real-time performance. Wang et al. [24] developed the Dynamic Distance–Intensity Outlier Removal (DDIOR) filter, which is an extension of DSOR by considering the distance and intensity values of snow noises on the basis of DSOR.

In the subsequent sections, four different approaches that have been adopted for LiDAR noise removal, including two general-purpose methods (i.e., ROR and SOR) and two state-of-the-art methods specifically designed to process LiDAR data (i.e., DROR and DSOR), are described in detail. It is worth noting that all the four methods are implemented in the "Experimental Study" section for cross comparison purposes.

- Statistical Outlier Removal (SOR)

The SOR filter [17] is a general-purpose filter that has been widely adopted to remove noises from point cloud data. The SOR filter first performs a k-nearest neighbor search for each point in the point cloud and then calculates the average distance to its neighboring points. The mean and standard deviation of the average distances are calculated to determine a global threshold defined as (1). Points whose distances to their k-nearest neighbors are larger than the threshold T_g will be classified as outliers. Because the SOR filter employs a global threshold for outlier detection, it does not consider the non-uniform distribution of LiDAR point cloud data.

$$T_g = \mu + (\sigma \times \beta) \tag{1}$$

where μ and σ denote the mean and standard deviation of the average distances from all points to their k-nearest neighbors; β is a multiplication factor.

- Radius Outlier Removal (ROR)

The ROR filter [17] is also a general-purpose filter designed to remove noises from point cloud data, which iterates over each point in a point cloud and calculates the number of neighboring points within a fixed search radius. If the total number of neighboring points is smaller than a threshold, then the point of interest is identified as an outlier. As mentioned by [8] and [11], the ROR filter showed deteriorated performance on snow removal because it does not consider the non-uniformity of the LiDAR data.

- Dynamic Radius Outlier Removal (DROR)

Proposed by Charron et al. [8], the DROR filter was extended from the ROR filter, which addresses the non-uniformity of LiDAR point cloud by dynamically adjusting the search radius in the ROR filter based on the point-to-sensor distance, as expressed in (2) [8]. In [8], the DROR filter was compared against the ROR filter on real-world LiDAR data corrupted by snowflakes and showed better performance on snow removal.

$$SR_p = \begin{cases} SR_{min} & r_p < SR_{min} \\ \beta \times r_p \times \alpha & otherwise \end{cases} \tag{2}$$

where SR_p denotes the search radius for the point p; r_p refers to the distance between p and the LiDAR sensor; SR_{min} is a minimum threshold for the search radius; α denotes the horizontal angular resolution of the LiDAR sensor; β is a multiplication factor.

- Dynamic Statistical Outlier Removal (DSOR)

The DSOR filter was extended from the SOR filter by employing a dynamic threshold on the average distances instead of a fixed threshold. The modification on the threshold can be expressed as (3) [11]. If the average distance between a point and its k-nearest neighbors is larger than T_d, it will be classified as an outlier. A comparative study was performed in [11] using real-world LiDAR data, which showed that the DSOR filter outperformed the SOR filter and DROR filter on snow removal.

$$T_d = T_g \times r \times d \tag{3}$$

where T_d denotes the dynamic threshold adjusted for each point; T_g denotes the global threshold defined in (1); r is a multiplication factor; d refers to the point-to-sensor distance.

III. Methodology

This section first introduces the LiDAR working mechanism and the characteristics of LiDAR data and snow noises, and then describes the technical details of the proposed LiDAR de-noising methodology. Subsequently, the physical meanings of the parameters involved in this methodology are elaborated through theoretical derivations, and their impacts on the de-noising performance are also discussed and assessed qualitatively.

A. LiDAR Working Mechanism

3D LiDAR data usually have a very high resolution along the horizontal direction but a relatively low resolution along the vertical direction. For example, for the Velodyne VLP-32C LiDAR sensor employed in this study, the horizontal angular resolution is 0.2°; in contrast, the angles between laser channels along the vertical direction range from 0.33° to 9.36°. Fig. 1 illustrates a schematic diagram of the sensor laser pattern for VLP-32C. This type of sensor consists of 32 laser

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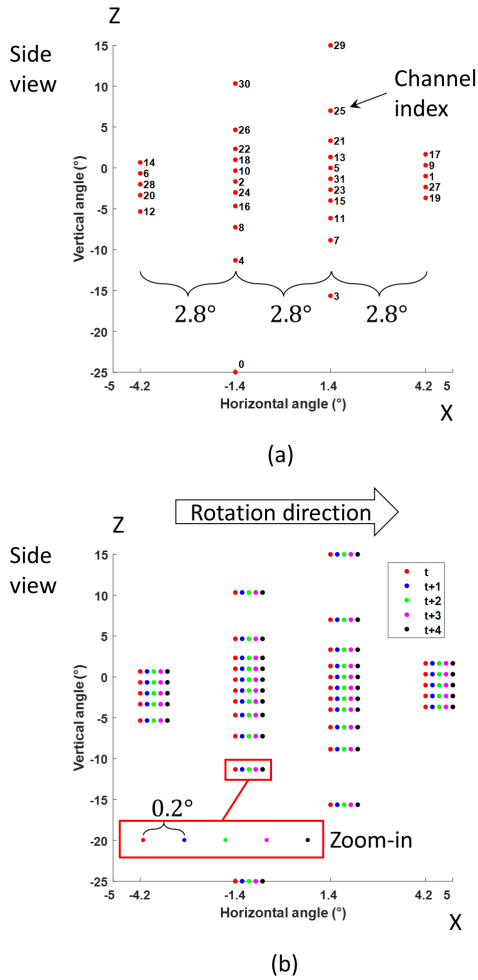

Fig. 1. Schematic diagram showing the sensor laser pattern: (a) a single firing; and (b) five consecutive firings.

channels, and each channel is a single 902-nm Infrared (IR) laser emitter and detector pair. Note that each channel has a unique and fixed elevation angle relative to the horizontal plane of the sensor, and its horizontal angle relative to the other laser channels is also fixed, as shown in Fig. 1 (a). This figure also indicates that the laser channels are arranged into four groups such that the channels in each group are inline vertically and each group has a horizontal angular offset (i.e., 2.8°) to the other adjacent groups. This sensor pattern keeps a widely separated angle between each two adjacent laser channels that are fired in pairs (e.g., channel 0 and channel 1, channel 2 and channel 3, etc.) to minimize IR crosswalk. When the IR laser emits a laser pulse, its time-of-shooting and direction are recorded. As the laser pulse travels through air and gets reflected by an object, the detector receives a portion of the reflected laser energy and records the time-of-acquisition. Thus, the distance between the laser and the object can be readily calculated based on the speed of laser pulse and the time difference.

During data acquisition, the LiDAR sensor rotates about its vertical axis with a constant spin rate (i.e., 600 RPM in this study) to obtain a 360° horizontal field-of-view (FOV). Fig. 1 (b) shows the sensor laser pattern after five consecutive

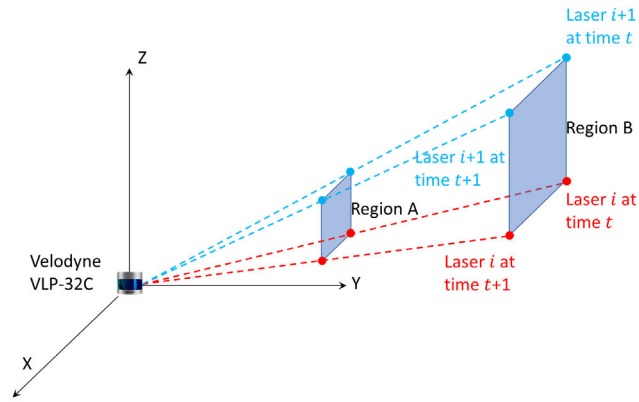

Fig. 2. The non-uniformity of LiDAR point cloud.

firings, as the sensor is self-rotating. Thus, the horizontal angular resolution can be calculated as (4).

$$\alpha = 360 \cdot \frac{\varphi}{60} \cdot t_0 = 0.2^\circ \tag{4}$$

where φ is the spin rate of the LiDAR sensor, equal to 600 RPM in this study; t_0 denotes the firing timing of the sensor, equal to 55.296 μs per firing cycle.

B. LiDAR Data Characteristics

Because laser lights travel straight, the density of the captured 3D point cloud reduces as the point-to-sensor distance increases. An example in Fig. 2 shows the scenario when two laser channels capture four points after two consecutive firings. It is obvious that the area of the rectangular region enclosed by the four laser pulses is dependent on the distance between the point and the laser. At a farther distance, with a larger region covered (i.e., Region B > Region A), the point density in that region will reduce accordingly.

The other contributing factors to the non-uniformity of LiDAR data include the irregular shapes of the captured objects, the existence of noises, the diversity of background, etc. Especially, for LiDAR data captured in snowy weather, because the tiny snow particles can easily backscatter laser beams, these snowflakes often present themselves in the point cloud as diffuse point clusters spreading around the sensor. As observed by [6], snow noises in LiDAR data usually follow a log-normal distribution, typically detectable within a close range (e.g., 10∼15 meters) to the sensor.

The unique characteristics of LiDAR data and snow noises determine that conventional methods for general-purpose noise removal are unequipped to effectively de-noise LiDAR data while preserving important edges and foreground features [8].

C. Proposed Methodology

- Dynamic search radius

In a 3D LiDAR point cloud, foreground objects such as vehicles are often shown as massive point clusters, with each point in close proximity to its neighboring points in the same cluster; in contrast, noises caused by snowflakes exist in the form of isolated points that are dispersing around the center of the

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point cloud (i.e., sensor location). Based on this observation, the proposed methodology, Dynamic Channel-wise Outlier Removal (DCOR) filter, identifies data outliers based on the distances from each data outlier to all the points in its vicinity: if the number of neighboring points within a specific distance (i.e., search radius) to the point of interest is smaller than a threshold, then that point is considered isolated and labeled as an outlier.

The term "dynamic" indicates that, different than existing general-purpose methods, the proposed methodology dynamically adjusts the search radius for each point based on its distance to the sensor upon calculating the number of neighboring points, to address the non-uniformity of LiDAR data.

- Channel-wise outlier removal

As can be inferred from the term "channel-wise", another unique feature of the proposed methodology is that it processes LiDAR data channel-by-channel, unlike the state-of-the-art methods (i.e., DSOR, DROR) which choose to filter the entire point cloud. In 3D LiDAR data, the captured point clouds are organized based on their laser channels, and the point cloud data in a specific laser channel share many common attributes, such as elevation angle. Processing LiDAR data in a channel-wise manner can reduce the data dimensionality and decouple the effects of snow noises along the vertical axis. With the channel-wise data, the noise removal task has been simplified, and the proposed methodology can now focus on distinguishing between the snow noises and foreground points only in that channel, based on their distinct geometric features. In addition to adopting the channel-wise data processing strategy, the proposed methodology differs from DROR and DSOR in the following aspects: *i*) the DROR filter requires to specify a minimum search radius for LiDAR points with a very small distance to sensor, as expressed in (2); in contrast, the proposed DCOR filter does not employ such a piecewise-linear design for the dynamic search radius; and *ii*) the DSOR filter removes snow outliers by comparing the mean distance from a point of interest to its k nearest neighbors against a distance threshold; on the contrary, the proposed DCOR filter determines if a point is an outlier by assigning a minimum threshold on the number of neighboring points within a predefined dynamic search radius.

The pseudocode for the proposed DCOR filter is provided as Algorithm 1. The distance factor f and the minimum number of neighboring points (i.e., a threshold) *MinPts* are the two control parameters involved in the proposed methodology. The physical meanings and impacts of these two parameters are described in the subsequent section.

It is worth noting that, in the "Experimental Study" section, a variant of the proposed methodology, referred to as DCOR-variant1, is also implemented for cross comparison purposes. The only difference between DCOR and DCOR-variant1 is that the latter adopts a fixed search radius. Thus, by comparing the de-noising performance between the proposed methodology and its variant, the impact from the use of a dynamic search radius can be demonstrated.

- Physical meanings of the parameters f and *MinPts*

Algorithm 1 Dynamic Channel-wise Outlier Removal

Input:
LiDAR point cloud (**P**) $= \{P_1, P_2, \ldots, P_n\}$; $n =$ number of laser channels
$f =$ distance factor
MinPts = minimum number of neighboring points
Output:
Outliers (**O**)
Filtered point cloud (**F**)

```
for P_i ∈ P do // Iterate over point clouds
    for p ∈ P_i do// Iterate over laser channels
        (x, y, z) ← coordinate of p
        distance ← sqrt(x^2 + y^2 + z^2)
        // Calculate the dynamic search radius
        SearchRadius ← distance × f
        // Search for the neighboring points
        k ← NeighboringPoints(p, SearchRadius)
        // Remove the snow noises
        if k < MinPts then
            add p to O
        else
            add p to F
        end if
    end for
end for
return O and F
```

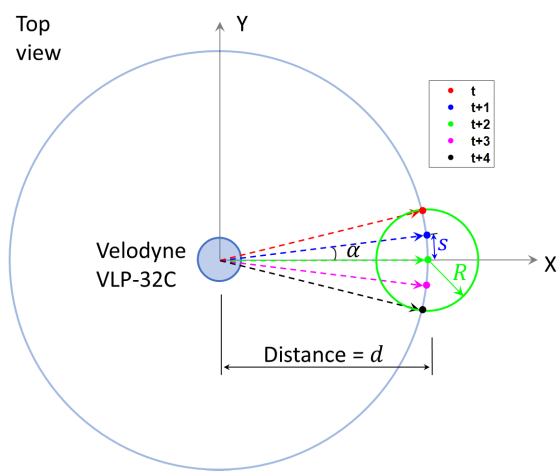

Fig. 3. Schematic diagram showing the captured points by a single laser channel through five consecutive firings.

Fig. 3 illustrates a schematic diagram of five points captured by a laser channel through consecutive firings. The distance between the point of interest, shown as a green dot in Fig. 3, and its nearest neighbor (shown as a blue dot) can be calculated as (5). In the meantime, it is worth noting that the subsequent derivation processes are based upon single-channel LiDAR data.

$$s = 2d \cdot \sin\left(\frac{\alpha}{2}\right) \tag{5}$$

where s is the Euclidean distance between the point of interest to its nearest neighboring point; d denotes the distance from

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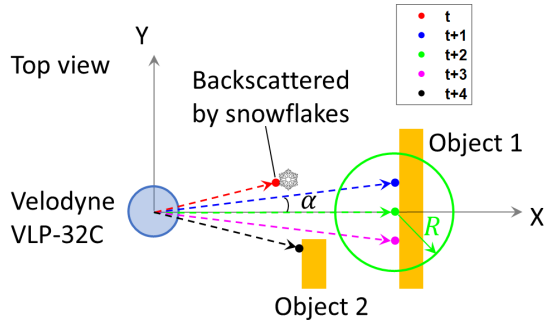

Fig. 4. Schematic diagram showing the captured points by a single laser channel through five consecutive firings, under snow noises and occlusion.

the point of interest to the sensor; α is the horizontal angular resolution, as expressed in (4).

If we employ s in (5) as the search radius for the point of interest, the number of neighboring points (including itself) within this radius is three, under the scenario depicted by Fig. 3. Similarly, we expand the range of the search radius by letting it cover at least NP neighboring points, and its expression can be formulated as (6).

$$R = 2d \cdot \sin\left(\frac{\mathrm{NP}-1}{2} \cdot \frac{\alpha}{2}\right) \tag{6}$$

where R refers to the minimum search radius to enclose NP points; NP denotes the number of neighboring points within the search radius.

From (6), we define the distance factor as the search radius divided by the distance from the point of interest to the sensor location, expressed as (7). This expression provides a theoretical basis for designing a search space for the optimal f value in the subsequent "Experimental Study" section. For example, in order to capture at least three points, the lower bound for the corresponding distance factor is calculated as 0.0035, based on (7). Meanwhile, (6) and (7) both show that the search radius specified for 3D LiDAR data is proportional to the point-to-sensor distance, thus proving the necessity of employing a dynamically adjusted search radius instead of a fixed one for neighborhood-based outlier removal.

$$f = \frac{R}{d} = 2\sin\left(\frac{(\mathrm{NP}-1)\cdot\alpha}{4}\right) \tag{7}$$

Note that the relationships in (6) and (7) only hold under "perfect" conditions without any snow noises. As illustrated in Fig. 4, with the existence of snowflakes and other objects, the number of points within the search radius is often much fewer than the theoretical value, which can be described by (8).

$$\mathrm{NP}_{\mathrm{actual}} \ll \frac{4\arcsin\left(\frac{R}{2d}\right)}{\alpha} + 1 \tag{8}$$

where $\mathrm{NP}_{\mathrm{actual}}$ denotes the actual number of neighboring points within a specific search radius when real-world challenges such as snow noises and occlusion exist.

The expression in (8) implies that, for neighborhood-based outlier removal as with the proposed methodology, the minimum threshold for the number of neighboring points (denoted as *MinPts*) is rather critical, and it remains a challenge to determine a proper threshold value to effectively distinguish between foreground points and snow outliers within a specific search radius. The *MinPts* value needs to be smaller than $\mathrm{NP}_{\mathrm{actual}}$, otherwise all the foreground points will be misidentified as outliers; on the other hand, if *MinPts* is much smaller than $\mathrm{NP}_{\mathrm{actual}}$, the algorithm then becomes less aggressive in detecting snow outliers, leading to more false negative detections.

Fig. 5. A LiDAR sensor installed on a traffic signal lighting pole at a road intersection.

The two parameters f and *MinPts* jointly control the performance of the proposed DCOR filter through coupled influences. That is, increasing the value of one parameter will essentially require to increase the value of the other accordingly, in order for the proposed methodology to reduce false positive/negative detections and operate with effectiveness and consistency. In the existing literature, there is no exact guideline on the design procedures for these two parameters, and their optimal values often need to be determined through extensive experiments. In Case I of the experimental study, we employ grid search, a straightforward yet effective design approach, to perform an exhaustive search for the joint optimal configuration on f and *MinPts* from a large search space.

IV. Experimental Study

This section first describes the data acquisition and ground truth generation processes, and then briefly introduces the performance metrics adopted for quantitative evaluation. Subsequently, in the "Experimental Results" subsection, two experimental cases are presented: in Case I, a parametric study is performed to obtain the optimal parameter values for the proposed methodology to achieve the highest accuracy in outlier detection; furthermore, in Case II, the proposed methodology is compared against some existing methodologies for cross comparison and performance demonstration purposes.

A. Data Acquisition

The LiDAR data adopted in the experimental study were acquired from a road intersection in Reno, Nevada, U.S. As illustrated in Fig. 5, a Velodyne VLP-32C sensor was

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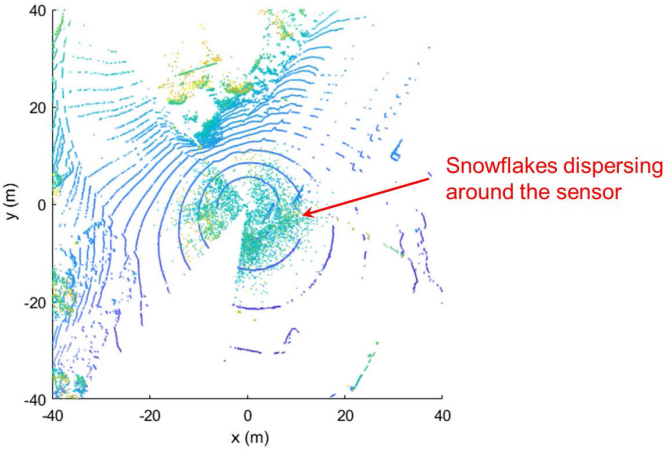

Fig. 6. An example of the acquired roadside LiDAR data corrupted by snow (bird-eye view).

installed on a traffic signal lighting pole at a corner of the road intersection for data collection. The LiDAR sensor features 32 channels of laser beams with a 40° FOV along the vertical direction and a 360° FOV along the horizontal direction. A data cabinet which houses the other equipment is also deployed at the site, containing a data processing computer, network communication devices, and external hard drives. Data acquisition was performed under severe snowy weather and resulted in 150 frames of LiDAR data for analysis. Fig. 6 provides an illustrative example of the captured LiDAR data corrupted by snow. As shown in Fig. 6, the LiDAR point cloud is contaminated with a massive amount of snow noises concentrated near the sensor location. Another observation from the captured LiDAR data is that the snow noises are predominantly scattered within 15 meters.

The captured LiDAR data were processed by using the DBSCAN-based LiDAR de-noising algorithm proposed by the authors in [19]. The obtained outlier detection results were further carefully examined and adjusted by a group of trained personnel to generate ground truth labels for the snow outliers.

Although the experimental study is performed using the Velodyne VLP-32C sensor, the proposed methodology can adapt to LiDAR data captured by different types of sensors (e.g., VLP-32C vs. VLP-16) by tuning the f and *MinPts* values accordingly, because the underlying mechanism of the proposed methodology stays unchanged, assuming the differences between the captured LiDAR data lie in their resolution, number of laser channels, elevation angle, etc. Meanwhile, the observations and research findings in this section based on the 32-channel LiDAR data can provide prior knowledge and insights for future similar studies using LiDAR data captured by other types of sensors as well.

B. Performance Evaluation Metrics

The precision-recall analysis [26] is adopted to provide a quantitative measure of the de-noising performance of the implemented methodologies. This analysis is comprised of three metrics, including the Precision, Recall, and F1 score, as defined in (9), (10), and (11), respectively. In these equations, *TP*, *FP*, and *FN* denote the number of true-positive detections, false-positive detections, and false-negative detections, respectively. Precision is defined as the number of true-positive detections divided by the total number of positive detections (i.e., *TP*+*FP*); Recall is defined as the number of true-positive detections divided by the total number of true objects (i.e., *TP*+*FN*). Precision reflects the ability of an algorithm to make the prediction results relevant to the true objects, while Recall evaluates the ability to classify all the true objects correctly. By taking the harmonic mean of Precision and Recall, the F1 score provides a comprehensive evaluation of the classification performance of an algorithm. In the experimental study, the F1 score is employed as the primary metric to quantitatively measure the accuracy of the adopted methodologies for snow removal.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{9}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{10}$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

C. Experimental Results

Two experimental cases are performed in this section. In Case I, a parametric study is conducted using the captured LiDAR data to investigate the optimal configuration of the distance factor f and the minimum number of neighboring points *MinPts*. Subsequently, in Case II, the proposed methodology is compared against some existing methodologies regarding their de-noising performance. The experiments are performed in MATLAB 2022a with the following computer specifications: CPU: Intel Core i7-8750H @2.20 GHz; RAM: 16 GB.

- Case I: A parametric study to determine the optimal parameter configuration on f and *MinPts*

In this subsection, a parametric study is performed through grid search to explore the optimal choice on the f and *MinPts* values, such that the proposed DCOR filter can achieve the highest F1 score on snow removal. The values for the distance factor f range from 0.005 to 0.1 with an incremental step of 0.005, as expressed in (12). Meanwhile, the values for the minimum number of neighboring points *MinPts* are odd numbers ranging from 3 to 21, as shown in (13). Thus, 200 combinations of f and *MinPts* values are considered in the experimental study. In Case I, each set of the 200 parameter combinations is assigned to the proposed methodology to evaluate the corresponding performance on filtering 150 frames of LiDAR data. For concision, the detailed statistics of the precision-recall metrics are not presented herein; instead, the Precision, Recall, and F1 score values (averaged over the 150 frames of LiDAR data) for the 200 subcases are illustrated in Fig. 7, Fig. 8, and Fig. 9, respectively, to provide a graphical representation on the performance with different parameter values.

$$f = 0.005ii \in \{1, 2, 3, \ldots, 20\} \tag{12}$$

$$MinPts \in \{3, 5, 7, \ldots, 21\} \tag{13}$$

In Fig. 7, Fig. 8, and Fig. 9, the horizontal x and y axes refer to the f and *MinPts* values, respectively, while the

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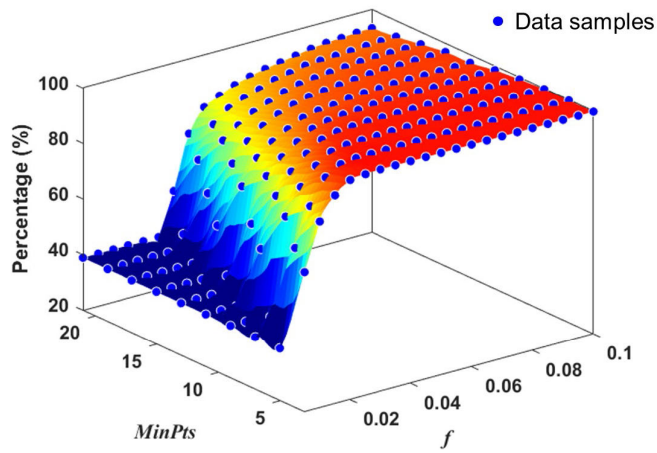

Fig. 7. Case I: Fit a surface to the Precision values.

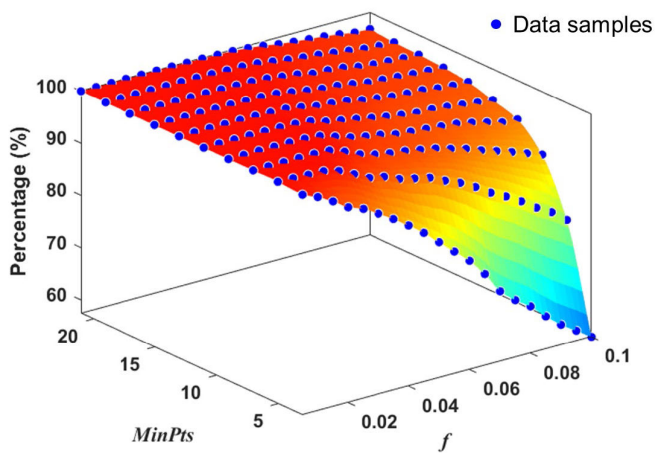

Fig. 8. Case I: Fit a surface to the Recall values.

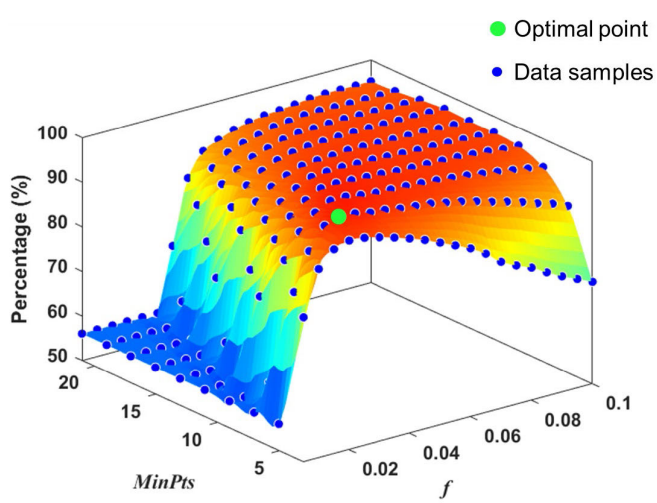

Fig. 9. Case I: Fit a surface to the F1 score values.

vertical axis refers to the percentage value of each metric. In these figures, the 200 data samples are shown as blue dots; meanwhile, a surface is fitted to each of the calculated metrics through cubic interpolation to help interpret the changing trend incurred by different parameter values.

Fig. 7 illustrates the average Precision value evaluated on the captured LiDAR point clouds under different parameter configurations. As can be observed, when the distance factor f is relatively small (e.g., 0.01), employing a large value for the *MinPts* threshold leads to more aggressive detection of outliers. The resulted Precision values by the proposed DCOR filter approach the ratio between the number of snow points and the total number of points in the investigated area (i.e., ≈40% in this study) because the algorithm tends to predict every point as an outlier, as illustrated by the deep blue region in Fig. 7. Meanwhile, increasing the search radius through enlarging the distance factor f makes the algorithm less aggressive, resulting in fewer false-positive detections and improved Precision values, as represented by the deep red region in the figure. Overall, it can be observed as a clear trend that a smaller *MinPts* value with a larger f value leads to a higher Precision value.

Similarly, the average Recall values and the fitted surface are plotted in Fig. 8. It can be observed that the distance factor f governs the performance of the Recall value. Under a fixed threshold on the minimum number of neighboring points (e.g., $MinPts = 3$), when the distance factor f increases, the proposed DCOR filter becomes less aggressive in classifying snow points as outliers due to a larger search radius, thus producing more false-negative detections (indicated by the reduced Recall value). Meanwhile, by increasing the value for the *MinPts* parameter, the proposed methodology tends to classify all the points as snow noises, leading to a 100% Recall value at the cost of misidentifying foreground points as noises, as shown by the deep red region in Fig. 8. Overall, the Recall value will increase with a smaller f value and a larger *MinPts* value.

Fig. 9 illustrates the average F1 score and the fitted surface. The F1 score is the harmonic mean of the Precision and Recall value, which provides an overall measure of the performance of the proposed methodology on snow removal. It can be clearly observed that the F1 score reaches the peak value when $MinPts = 5$ and $f = 0.025$, as indicated by the green dot in Fig. 9. Thus, through Case I, the optimal parameter set for the proposed methodology to achieve the highest accuracy in outlier detection has been determined.

- Case II: Cross comparisons on the outlier detection performance by the implemented methodologies

The proposed methodology is further compared against the following methodologies: DCOR-variant1, ROR, DROR, SOR, and DSOR. Note that the DCOR-variant1 filter is the same as the proposed methodology except that it does not adjust the search radius based on different point-to-sensor distances. The intention of implementing the DCOR-variant1 filter for cross comparisons is to reveal the impact of using a dynamic search radius to account for the non-uniformity of LiDAR data.

In total, six methodologies are implemented in Case II and applied to the captured LiDAR data for snow removal. The following parameters are adopted for these methodologies: *i*) DCOR: $MinPts = 5$, $f = 0.025$; *ii*) DCOR-variant1: $MinPts = 5$, search radius = 0.5 meters; *iii*) ROR: $MinPts = 5$, search radius = 0.5 meters; *iv*) DROR:

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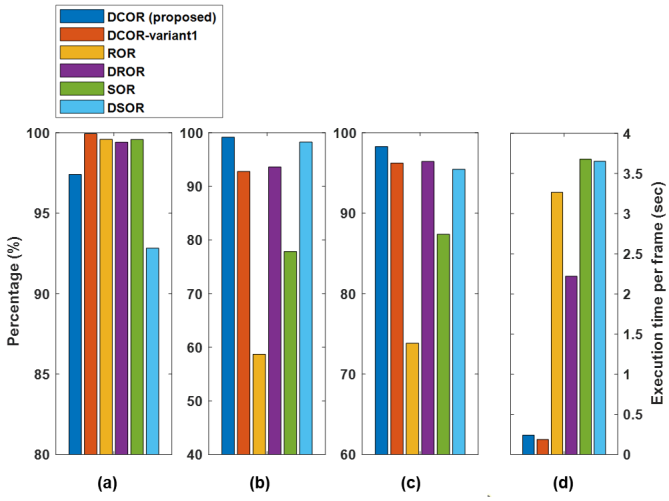

Fig. 10. Case II: Performance metrics by the implemented methodologies: (a) Precision; (b) Recall; (c) F1 score; and (d) execution time per frame.

TABLE I
CASE II: PERFORMANCE METRICS OF THE IMPLEMENTED METHODOLOGIES (AVERAGED OVER 150 FRAMES OF LiDAR DATA)

Algorithm	Precision (%)	Recall (%)	F1 score (%)	Execution time per frame (sec)
DCOR	97.4	99.2	98.3	0.242
DCOR-variant1	99.9	92.8	96.2	0.188
ROR	99.6	58.7	73.8	3.266
DROR	99.4	93.6	96.4	2.221
SOR	99.6	77.8	87.4	3.678
DSOR	92.8	98.2	95.4	3.652

$SRmin = 0.05$ meters, $\beta = 0.1$, $\alpha = 0.333°$, $MinPts = 5$; *v*) SOR: $\beta = 0.1$, $MinPts = 5$; and *vi*) DSOR: $\beta = 0.1$, $MinPts = 5$, $r = 0.05$. Note that for each of these methodologies except the proposed one, the parameter configuration has been validated through prior knowledge and preliminary studies, and the optimal choice on their values is not discussed herein for concision.

The following performance metrics, including the average Precision, Recall, F1 score, and execution time per frame, are illustrated in Fig. 10, with detailed performance metrics tabulated in TABLE I. Several observations can be made from Fig. 10 and TABLE I as follows:

1) Proposed DCOR vs. DCOR-variant1: When the point-to-sensor distance is less than 20 meters, the fixed search radius (0.5 meters) adopted in the DCOR-variant1 filter is much larger than the dynamic search radius (distance factor = 0.025) employed by the proposed DCOR filter. As a result, the differences between DCOR-variant1 and DCOR in the Precision and Recall values are 2.5% and −6.4%, respectively. Such observations are consistent with the findings from Fig. 7 and Fig. 8. Meanwhile, judging from the F1 score, the proposed DCOR filter with an adaptive search radius shows better performance than DCOR-variant1, where the improvement in the F1 score is 2.1%.

2) Proposed DCOR vs. DROR/DSOR: According to TABLE I, the F1 score values for DCOR, DROR, and DSOR are 98.3%, 96.4%, and 95.4%, respectively. Compared to the state-of-the-art methodologies, the proposed DCOR filter yields the highest F1 score on snow removal, with an improvement of 1.9% and 2.9%, respectively. Thus, through Case II, the accuracy of the proposed methodology on snow removal has been validated through cross comparisons with the two state-of-the-art methodologies.

3) DCOR-variant1 vs. ROR: The significant difference between DCOR-variant1 and ROR lies in that the former processes the LiDAR data in a channel-wise manner to decouple the snow noises along the vertical axis of the 3D space for a more effective outlier detection. In contrast, the latter directly filters the entire point cloud. As shown in Fig. 10 and TABLE I, DCOR-variant1 outperforms ROR in every metric, where the improvements in the Precision, Recall, and F1 score values are 0.3%, 34.1%, and 22.4%, respectively. This result suggests that processing LiDAR data channel-by-channel will significantly boost the performance in correctly identifying all the snow noises, as indicated by the higher Recall value.

4) DROR vs. ROR: Using a dynamic search radius for outlier detection, the DROR filter shows improved results compared to ROR. From TABLE I, the differences between DROR and ROR in the Precision, Recall, and F1 score values are −0.2%, 34.9%, and 22.6%.

5) DSOR vs. SOR: The comparison between DSOR and SOR shows the differences in the Precision, Recall, and F1 score values are −6.8%, 20.4%, and 8.0%, respectively. By employing a dynamic threshold that changes according to the point-to-sensor distance rather than a fixed threshold [see (3)], the DSOR filter yields higher values on the Recall and F1 score, showing better overall performance on snow removal.

6) Comparison on the Efficiency of the Implemented Methodologies: from Fig. 10, the proposed DCOR filter and its variant both yield drastically reduced execution time compared to the other implemented methodologies. For example, the execution time per frame for DCOR, DROR, and DSOR are 0.242, 2.221, and 3.652 secs, respectively, indicating that the proposed methodology has much higher efficiency in snow removal. The reason behind such a significant improvement in the execution efficiency is that the proposed methodology chooses to decouple the effect of snow noises and reduce the data dimensionality by processing data channel-by-channel. Thus, with a much smaller search space in each channel, the computational efforts needed for nearest-neighbor searches are effectively reduced.

Furthermore, an illustrative example is provided in Fig. 11 to qualitatively evaluate the performance of the implemented methodologies on snow removal. Fig. 11 (a) shows the raw LiDAR point cloud corrupted by heavy snow, while Fig. 11 (b-g) demonstrates the filtered point cloud by DCOR, DCOR-variant1, ROR, DROR, SOR, and DSOR, respectively. Judging from the graphical results in Fig. 11 (b), (e), and (g),

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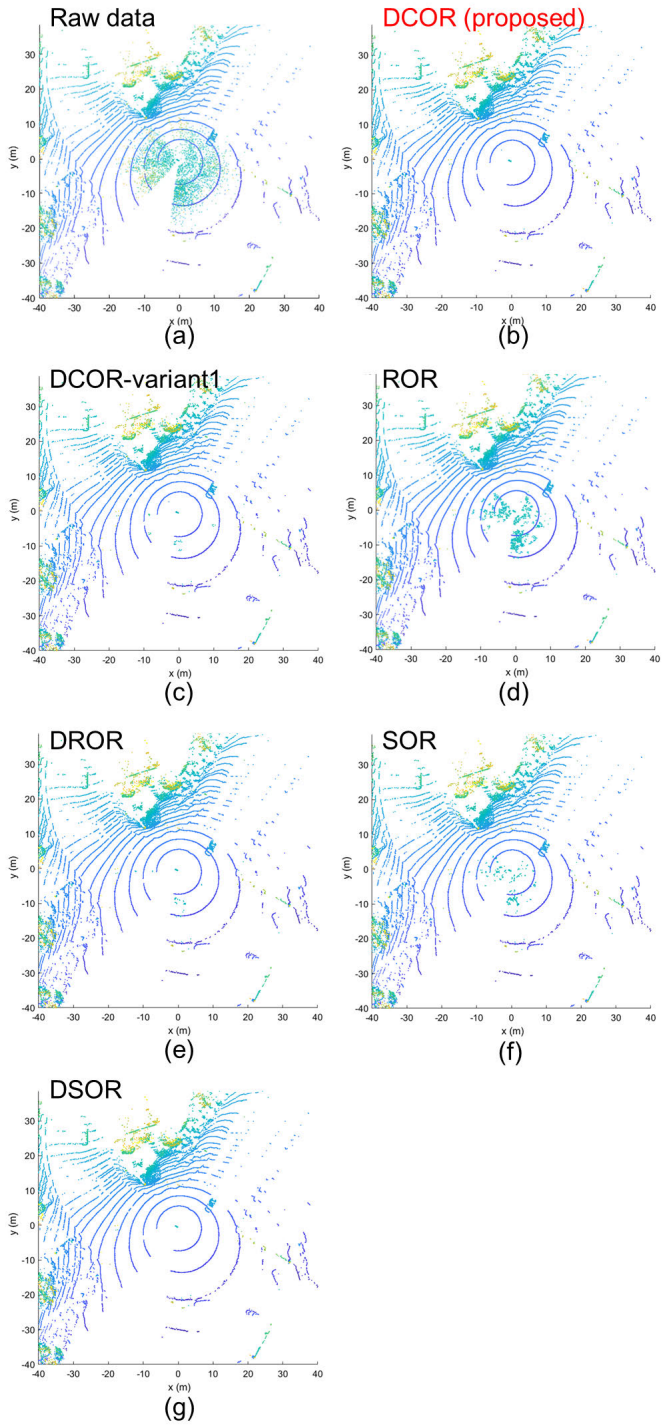

Fig. 11. Case II: An example of the de-noising result: (a) raw point cloud with snow noises; (b) filtered point cloud by DCOR; (c) DCOR-variant1; (d) ROR; (e) DROR; (f) SOR; and (g) DSOR.

DCOR, DROR, and DSOR yield better performance on snow removal than the other methodologies. On the other hand, some remnants of the snow noises can be seen in the filtered results, especially by ROR [Fig. 11 (d)] and SOR [Fig. 11 (f)], because these two types of filters are general-purpose filters that do not consider the non-uniformity of LiDAR data.

Fig. 12 further evaluates the performance of DCOR, DROR, and DSOR on eliminating snow noises at difference distances.

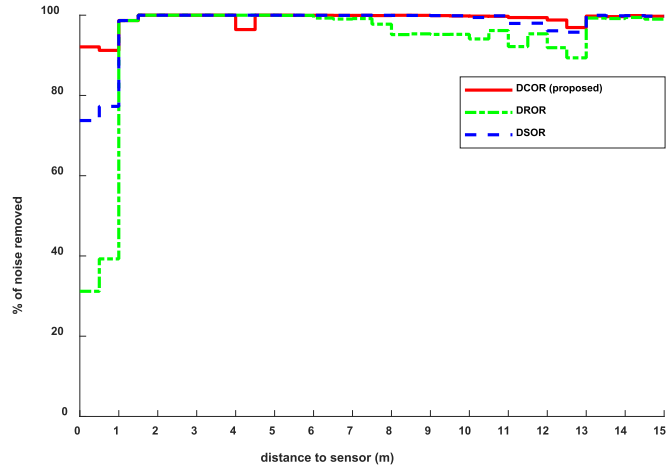

Fig. 12. Case II: Percentage of removed snow noises as a function of the distance to sensor.

In Fig. 12, the horizontal axis represents the distance between the snow noises and the sensor, and the vertical axis refers to the percentage of removed snow noises, which is the same as Recall. It can be seen that the proposed DCOR filter captures more snow noises than the other two in the range of 0∼1 meters. As the distance to the sensor keeps increasing, DCOR shows similar performance to DSOR, and they both outperform DROR in the range of 7∼13 meters. Such observations are consistent with the results on the Recall metric (see TABLE I).

V. Conclusion

Severe weather conditions such as heavy fog and snow can often reduce the quality of LiDAR data because laser beams can be easily backscattered by fog and snow particles. Tiny snow particles are usually shown as diffuse, solid objects scattered around the sensor location in a LiDAR point cloud. Such disturbances must be effectively eliminated before the LiDAR data captured under severe weather can be adopted for the subsequent traffic object recognition, safety assessment, and decision-making tasks.

This study proposes a novel methodology called Dynamic Channel-wise Outlier Removal (DCOR) to de-noise LiDAR data corrupted by snow. The proposed methodology iterates over the LiDAR data based on different laser channels and marks a point of interest in a channel as an outlier if the number of neighboring points in the same channel (within a dynamic search radius) is fewer than a threshold. Unlike the existing LiDAR de-noising methodologies, The proposed methodology processes LiDAR data in a channel-wise manner, to reduce the data dimensionality and decouple the snow effects along the vertical axis; furthermore, upon searching for neighboring points, the proposed methodology adopts a dynamically adjusted search radius which is proportional to the point-to-sensor distance, to account for the varying point density that decreases as the distance to sensor increases.

Two control parameters, namely the distance factor f and the minimum number of neighboring points $MinPts$, are

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involved in the proposed methodology. The first parameter f is utilized to modify the search radius for each point of interest, and the second parameter *MinPts* sets a minimum threshold to identify outliers based on the number of nearest neighbors. In the "Methodology" section, their physical meanings and impacts to the de-noising performance of the proposed methodology are elaborated and assessed qualitatively. These two parameters jointly control the performance of the proposed DCOR filter through coupled influences: increasing the value of one parameter will essentially require to increase the value of the other accordingly, in order for the proposed methodology to reduce false positive/negative detections and operate with effectiveness and consistency.

Data acquisition was performed at a road intersection in Reno, Nevada, U.S., resulting in 150 frames of LiDAR data with snow noises for analysis. In Case I of the experimental study, a parametric study is conducted to investigate the optimal choice of the distance factor f and the minimum number of neighboring points *MinPts*, such that the proposed methodology can yield the highest accuracy in snow removal. A large search space consisting of 200 combinations of f and *MinPts* values is constructed to explore the joint optimal parameter configuration through grid search. It has been observed as a general trend that a larger value for f expands the search radius, and thus, the DCOR filter tends to detect more snow noises as false negatives (i.e., reduced Recall value); a larger value for *MinPts* makes the DCOR filter more aggressive in outlier detection, leading to more false positives (i.e., reduced Precision value). Through Case I, the optimal values for f and *MinPts* that lead to the highest F1 score are 0.025 and 5, respectively.

In Case II, the proposed DCOR filter is compared against some other methodologies, including DCOR-variant1, ROR, DROR, SOR, and DSOR, regarding their performance on snow removal. Note that the only difference between DCOR and DCOR-variant1 is that the former employs a dynamic search radius while the latter utilizes a fixed search radius. The experimental results show that the proposed DCOR filter outperforms its variant with a 2% increase in the F1 score by leveraging a dynamic search radius that considers the non-uniformity of LiDAR data. Meanwhile, the comparison between DCOR-variant1 and ROR demonstrates that the strategy of processing LiDAR data in a channel-wise manner can significantly boost the performance with an increase of 4.7% in the F1 score, where ROR processes the entire point cloud as opposed to DCOR-variant1. Moreover, through the cross comparisons in Case II, it has been demonstrated that the proposed DCOR filter outperforms the state-of-the-art methodologies, including DROR and DSOR, in both accuracy and efficiency, where the average F1 score is 98.3%, 96.4%, and 95.4%, and the execution time per frame is 0.242 secs, 2.221 secs, and 3.652 secs, respectively.

In the experimental study, the optimal values for the two parameters are obtained through grid search, which is a straightforward but time-consuming approach. Thus, further research efforts need to be devoted to developing more effective parameter tuning strategies to improve the efficiency and robustness of the proposed methodology.

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